Internet Appendix

Ask BERT: How Regulatory Disclosure of Transition and Physical Climate Risks affects the CDS Term Structure

July 1, 2022

Abstract

In this internet appendix, we provide a description of the CDS data used in our study. In addition, we provide a short and intuitive description of the BERT algorithm.

1 A brief introduction to BERT

In this appendix, we provide some intuition on how BERT works. A full description of all the aspects of this algorithm is far beyond the scope of this paper. Hence, we can only scratch at the surface and refer the interested reader to the original papers. BERT, as introduced by Devlin et al. (2019), is a new language representation model. Unlike Peters et al. (2018),[1] BERT pre-trains deep bidirectional representations from an unlabeled text by jointly conditioning on both left and right context in all layers. A pretrained BERT can be fine-tuned with just one additional output layer.

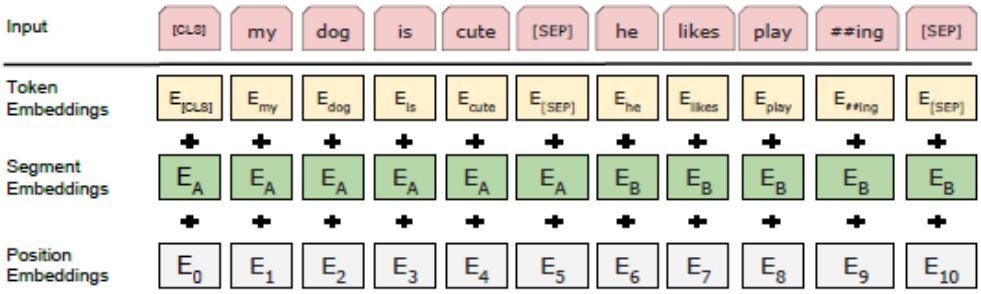
1.1 Pretraining BERT

For pretraining, BERT uses BookCorpus as data input, a dataset consisting of 11,038 unpublished books from 16 different genres and 2,500 million words from text passages of English Wikipedia. Indeed, it is critical to use a document-level corpus rather than a shuffled sentence-level corpus to extract long contiguous sequences. A striking feature of BERT is that one can start from the

[1] Peters et al. (2018) uses shallow concatenation of independently trained left-to-right and right-to-left language models.

Figure 1. BERT input representation

The input embeddings are the sum of the token embeddings, the segmentation embeddings, and the position embeddings. This figure corresponds to Figure 2 of Devlin et al. (2019).



pretrained BERT and then fine-tune BERT for specific tasks with less training efforts and data required.

As mentioned above, BERT relies on the Transformer, which is the attention mechanism that learns contextual relationships between words in a text. A Transformer consists of an encoder to read the input, and of a decoder to produce the output for the task. As illustrated in Figure 1, the input for BERT is a sequence of tokens with a specific structure, namely a token embedding, a segment embedding, and a positional embedding. The token embedding consists of a [CLS] token at the beginning (CLS stands for classification) and a [SEP] token between sentences (SEP stands for separation). The segment embedding is a marker indicating sentence A or sentence B so that the encoder can distinguish between sentences. Finally, a positional embedding is added to each token to indicate its position in the sentence.

1.1.1 Masked Language Model (MLM)

BERT alleviates the unidirectionality constraint by using a "masked language model" (MLM) pre-training objective. The MLM randomly masks some of the tokens from the input, and the goal is to predict the original index of the masked token in the vocabulary, only based on its context. MLM enables the representation to fuse the left and right context, which allows us to train a deep bidirectional transformer.

In BERT, training the language model is done by predicting 15% of the tokens in the input that were randomly picked. These tokens are pre-processed as follows – 80% are replaced with a "[MASK]" token, 10% with a random word, and 10% use the original word.[2] Since the Transformer encoder does not know which words it needs to predict or which have been replaced by random words, it must keep a distributional contextual representation of every input token. Hence, MLM converges slower than left-to-right LMs.

1.1.2 Next Sentence Prediction (NSP)

In addition to MLM, BERT solves a "next sentence prediction" task that jointly pre-trains text-pair representations. This task is implemented such that, when choosing the sentences A and B for each pretraining example, the probability that B is the actual next sentence that follows A is $1/2$, and with probability $1/2$ it is a random sentence from the text corpus.

Figure 1 illustrates the input structure required to help BERT distinguish between the two sentences in training. BERT uses the entire input sequence to predict if the second sentence is indeed connected to the first. The [CLS] token output from BERT is transformed into a 2×1 shaped vector using a simple classification layer (learned matrices of weights and biases). Then, the probability of sentence B being the next sentence of A is calculated using softmax.

When training the BERT model, MLM and Next Sentence Prediction are trained together, to minimize the combined loss function of the two strategies. There are four different types of pre-trained versions of BERT available, depending on the scale of the model and architecture:

- BERT-Base (cased and uncased): 12-layer, 768-hidden-nodes, 12-attention-heads, 110M parameters.

- BERT-Large (cased and uncased): 24-layer, 1024-hidden-nodes, 16-attention-heads, 340M parameters

[2]If one uses [MASK] 100% of the time the model would not necessarily produce good token representations for non-masked words. The non-masked tokens were still used for context, but the model was optimized for predicting masked words. If one used [MASK] 90% of the time and random words 10% of the time, this would teach the model that the observed word is never correct. If one used [MASK] 90% of the time and kept the same word 10% of the time, then the model could trivially copy the non-contextual embedding.

For our application, we rely on the BERT-Base uncased model.

1.2 Visualizing BERT's attention

In this section, we rely on Vig (2019) to visualize the inner working of BERT and focus on text embeddings.[3] The Transformer underlying BERT is a sequence model that replaces traditional recurrent architectures with an attention-based approach. Visualizing attention helps to interpret a model by showing how the model assigns weights to different input elements.

BERT learns multiple attention mechanisms. These mechanisms are called 'heads' and they work in parallel, i.e., independently of each other. Multi-head attention allows Bert to capture a much broader range of relationships between words. Besides, the BERT model is equipped with multiple layers of attention, which depend serially on each other. With this repeated composition of word embeddings, BERT can generate very rich representations as it gets to the deepest layers of the model. In the BERT-base, we have 12 layers and 12 attention heads.

To exemplify the attention mechanism for some text relating to climate change, we use the following two sentences.

Sentence A:

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The increase in natural disasters has adverse effects on our properties.
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Sentence B:

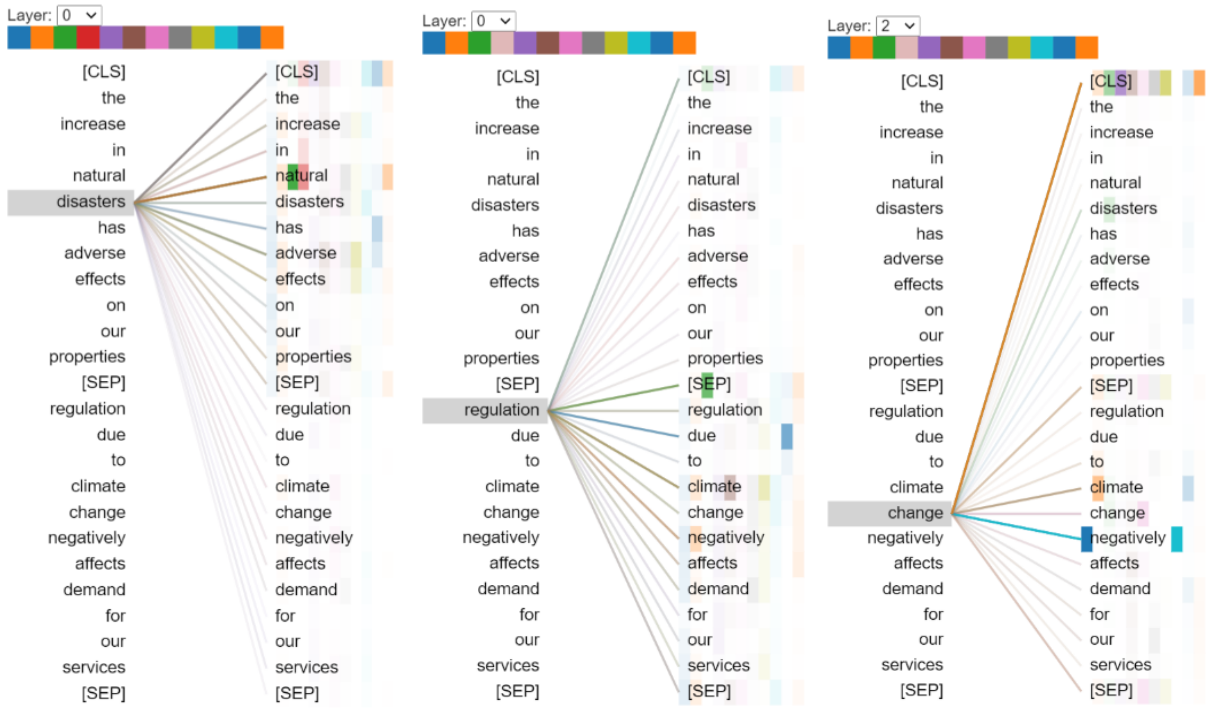
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Regulation due to climate change negatively affects demand for our services.
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In Figure 2, we represent different attention patterns for sentences A and B. The co-reference resolution suggests that BERT, still not fine-tuned for a climate-specific task, already relates 'disaster' to 'natural,' having 'adverse effects,' and it relates 'regulation' to 'climate change' and 'negatively affects.' Furthermore, on a different attention layer, BERT understands 'change' in its context, i.e., it relates it to 'climate' and 'negatively.' Besides, it relates 'change' from sentence B in the context of climate change to 'disaster' in sentence A.

[3]For words to be processed, they need some form of numeric representation that models can use in their calculation. A prominent example is word2vec, but - as noted above - these context-free methods are not well suited for language understanding.

Figure 2. Attention-head view for BERT

The figure represents different attention patterns for sentences A and B. Attention is represented as lines connecting the tokens that are attending (left) with the tokens being attended to (right). Colors identify the corresponding attention head(s), while line weight reflects the attention score.



BERT uses attention to compute a weighted average over word embeddings. A compatibility function assigns a score to each pair of words, allowing the computation of the attention weights. BERT first assigns to each word a query vector and a key vector. The dot product of these two vectors is then defined as the compatibility score. The final score is obtained by normalizing with the softmax function. Key and query vectors are the output from the previous layer and are computed dynamically based on the input to the model. For more details, we refer to Devlin et al. (2019).

To visualize how the query and key vectors are used to compute attention, Figure 3 traces the computation of attention from the selected word, in this case 'climate.' As an example, we take layer 2 and head 2, and show how attention from 'climate' on the left is directed to the complete sequence of words on the right. The column 'Query q' represents the 64-element query vector of the token paying attention. 'Key k' is the 64-element key vector of each token receiving attention. The element-wise product shows how individual elements in the query and key vectors contribute to

Figure 3. Neuron view

Neuron view for layer 2 and head 2, with the token 'climate' selected. Positive values are colored blue and negative values orange, with color intensity representing magnitude. Connecting lines indicate the strength of attention between the connected words.



the dot product. The dot product of the query vector and each key vector enters then the softmax, which eventually equals the attention received by the corresponding token.

For Figure 3, we have intentionally chosen layer 2 and head 2, as this choice exemplifies and nicely illustrates how BERT relates words, which are relevant for our climate project. The token 'climate' puts much weight on the [CLS] token. However, this weight is purely due to the architecture of the algorithm, and from the element-wise product, it seems that BERT only uses a few neurons to identify the position of this token. Also, BERT focuses on the next word, i.e., 'change,' which makes perfect sense as the next words are often the most relevant for understanding a word's meaning. However, BERT also relates words to 'climate' from the previous sentence, in particular the word

'disaster.' Moreover, attention is given to 'increase,' 'natural,' 'adverse,' and effects, which makes perfect sense in the context related to climate change.

1.3 BERT's sentence embeddings

Word embeddings are models to generate numeric vector representations for words. In contrast to context-free methods like word2vec, which produces 300 floats between 0 and 1 for every word, BERT offers context-dependent embeddings not only for words but for whole sentences. Hence, it is instructive to see whether BERT finds climate-related sentences similar to each other and whether we can intuitively visualize this property.

For that purpose, we use the uniform manifold approximation (UMAP), introduced by McInnes et al. (2018), to visualize the relations between sentences. UMAP is a novel manifold learning technique for dimension reduction. It constructs a high-dimensional graph representing the data and optimizes a low-dimensional graph to be as structurally similar as possible. We use UMAP to exemplify BERT's sentence embeddings in a three-dimensional space. In particular, we visualize pre-trained BERT embeddings of a set of sentences selected from the 10-K filings in Tensorflow's Tensorboard Embedding Projector.

Figure 4, plots a set of sentences based on BERT's embeddings in three dimensions determined by UMAP. To explore whether these sentences are intuitively close to each other, we choose the sentence:

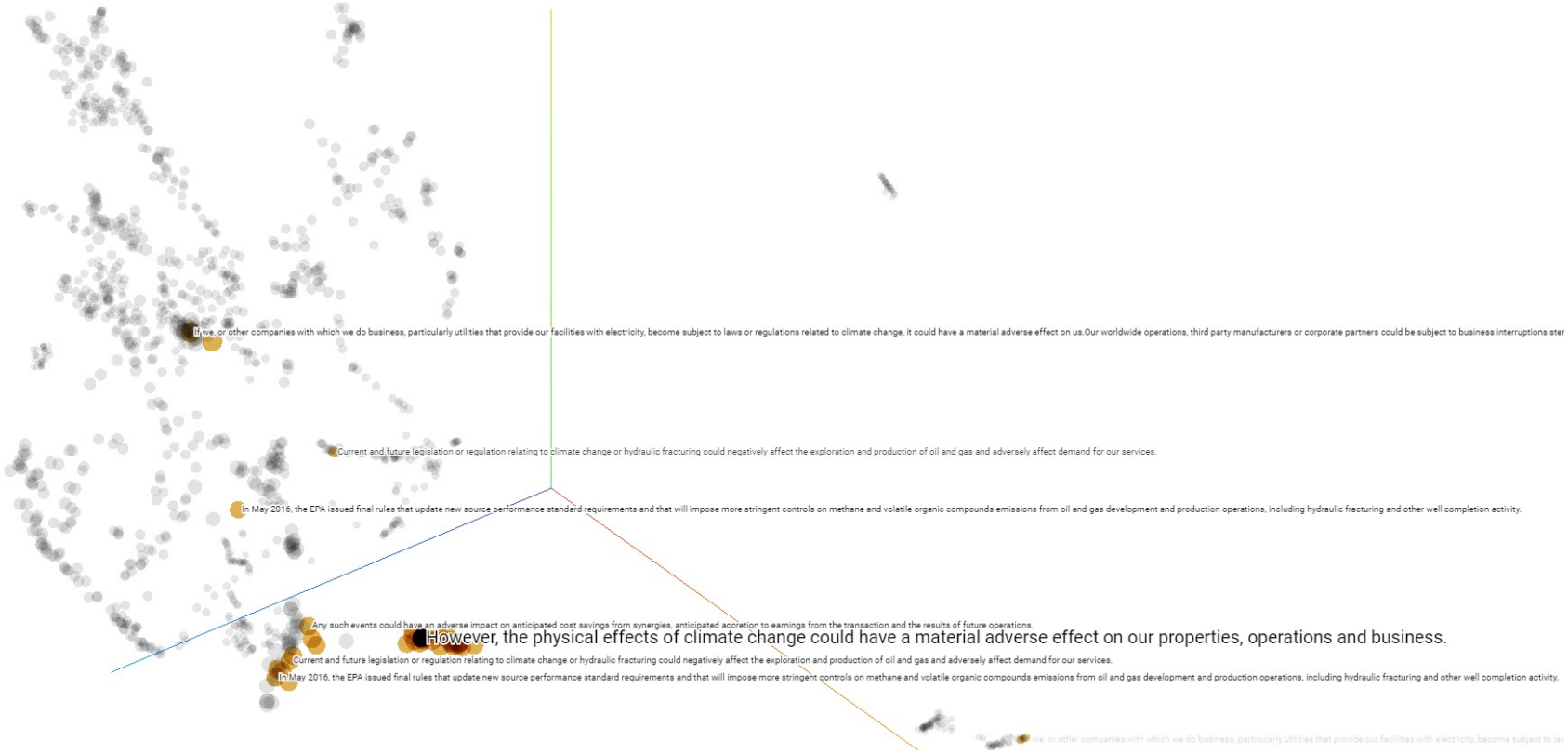
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However, the physical effects of climate change could have
a material adverse effect on our properties, operations, and
business.
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as a starting point, and we plot additional sentences that are similar. Interestingly, we observe that BERT determines sentences like:

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Current and future legislation or regulation relating to
climate change or hydraulic fracturing could negatively affect
the exploration and production of oil and gas and adversely
affect demand for our services.
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Figure 4. BERT's sentence embedding

This figure visualizes pre-trained BERT embeddings in Tensorflow's Tensorboard Embedding Projector. Tokens are represented as dots in a three dimensions determined with UMAP. Colored dots represent the nearest neighbors to the selected sentence.



being close to the first sentence mentioned above. We recall that the embeddings in Figure 4 are still based on a BERT model that has not yet been fine-tuned for climate-related tasks. However, as we observe, the pre-trained BERT interprets and connects important aspects of words and sentences, showing the potential of context-based methods. By fine-tuning BERT to a specific task, we can unleash its full potential.

1.4 Fine-tuning BERT

For a wide variety of language tasks, we can add one or more small layers to the pre-trained BERT model. This step is called fine-tuning and has its root in semi-supervised sequence learning (Dai and Le, 2015) and ULMFiT (Howard and Ruder, 2018). The fine-tuning step starts from the pre-trained BERT parameters and simultaneously updates BERT and trains the additional task-specific layer. Therefore, compared to pre-training, fine-tuning is relatively inexpensive and requires fewer data.

For our application, we are mainly interested in classification tasks. For each of these tasks, we can plug in the task-specific inputs and outputs into BERT and fine-tune all the parameters end-to-end. Regarding the inputs, BERT accepts pairs of text, which, in our case, is just the underlying text and an empty string pair. For the output of our classification tasks, only the [CLS] representation is taken and fed into the classification layers on the top.

For our classification tasks, we add two fully connected layers, both of size 128 on top of the output of the [CLS] token from BERT. We fine-tune BERT in a multi-task learning framework for TP and CN classifications. Multi-task learning is an effective approach to share the knowledge obtained from several related supervised tasks. Specifically, we let the TP and CN tasks share the BERT layers, while the classification layers are not shared. Hence, the BERT output is of higher generalization power to fit the needs of both tasks.

To select hyperparameters and train our model, we rely on our training data, which we obtained as follows. First, we use the sample reports provided in the TCFD (2019) guidelines, giving us almost a thousand examples for the CN and TP tasks. Second, we further collect random sentences that are not climate-related to build the initial dataset. We then feed the data to our model and train the model for a few epochs. Next, we run the trained model on the 10-K dataset, and we collect the

most confusing examples of the current model. We consider an example confusing when predicted probabilities for different classes are similar. We let human annotators review the confusing examples for labeling together with some confident examples. Adding the updated confident examples and the most confusing examples helps us to keep the probability distribution of the data not disturbed too much by the confusing examples. We perform this process for several rounds and end up with 3,192 data points. Given this data, we split our training set to use roughly 80% for training, 10% for validation and testing. The training set is then up-sampled to balance different classes. Since fine-tuning is typically fast, we run an exhaustive search over the grids of hyper-parameters and choose the model that performs best on the validation set. We choose to use a batch size of 16 and fine-tune for ten epochs over the data for the TP task. The fine-tuning learning rate is set as 1e-5. The dropout probability is kept at 0.1.

In Table 1, we list a few examples of how our fine-tuned BERT model classifies text passages into transition and physical risk, or into a category labeled as not related to climate risk. Given these examples, we are confident that by relying on BERT we can identify text from the 10-K filings that are related to climate risk. To obtain our final climate-risk score, we use the probabilities to determine to which category a set of sentences belongs. Adding these scores and normalizing with the total number of sentence pairs in Item 1.A gives us the total score of a given company. We use this as a proxy for climate-risk disclosure. In an earlier version, we also used some sentiment analysis on the climate-relevant text. However, this procedure did not give us much improvement. We believe that the reason for that finding is that the text in Item 1.A is already capturing the risk aspect, and also the forward-looking aspect. Hence, the formal structure of 10-K filings allows us to stay with a more straightforward approach to determine the climate-risk score. Such a structure is not present in voluntary reports, making it much harder to analyze using NLP tools. We leave this problem for future research.

2 CDS data collection

CDS spreads are retrieved from Thomson Reuters Datastream using the Datastream Excel add-in. We have located all CDS spreads through the "find series" option in excel and searched for CDS

Table 1. BERT-annotated text sequences

The table shows some examples of text sequences that were classified either as 'physical' or 'transition' climate risk, or as no climate-risk related text. For the classification task, we set the maximal number of tokens equal to 64 and we require BERT to classify two sentences to provide enough context.

		P(Classification)		
Cpy ID	Text passages related to transitional climate risk	phys	trans	other
47801	Compliance with and changes in various environmental requirements and environmental risks applicable to our industry may adversely affect our results of operations and financial condition. Occasionally, we may be required to clean up or take certain remediation action with regard to sites we formerly used in our operations.	0.0131	0.9758	0.0112
392659	Compliance with these foreign and domestic laws and related regulations is an ongoing process that is not expected to have a material effect on our capital expenditures, earnings or competitive position. Environmental concerns are, however, inherent in most major agricultural operations, including those conducted by us, and there can be no assurance that the cost of compliance with environmental laws and regulations will not be material.	0.0140	0.9742	0.0117
727425	In the future, we may incur increased costs and additional charges associated with environmental compliance and cleanup projects necessitated by the identification of new waste sites, the impact of new environmental laws and regulatory standards, or the availability of new technologies. Compliance with Federal, state and local environmental laws and regulations in the future may require a material increase in our capital expenditures and could adversely affect our earnings and competitive position.	0.0130	0.9758	0.0113
668070	Given these uncertainties, we currently estimate that capital expenditures for environmental purposes, excluding expenditures related to the Maximum Achievable Control Technology (or MACT) programs and landfill closures discussed below, will be $9 million in 2010, $6 million in 2011, and $6 million in 2012. The estimated expenditures could be significantly higher if more stringent laws and regulations are implemented.	0.0140	0.9745	0.0115
865496	Some European countries have implemented or are considering other initiatives for reducing CO2 vehicle emissions, including fiscal measures and CO2 labeling. For example, the United Kingdom, France, Germany, Spain, Portugal, and the Netherlands, among others, have introduced taxation based on CO2 emissions.	0.0131	0.9758	0.0111
Cpy ID	Text passages related to pyhsical climate risk	phys	trans	other
51378	For example, in 2011, several Midwest states, including Iowa, Indiana, and Ohio, recorded the wettest April in more than a century. These record levels of rainfall led to the flooding of the Mississippi River in April and May, restricting production and shipments in several of the Company's south central operations.	0.9823	0.0092	0.0085
962117	We Face Risks from Unpredictable Catastrophic Events. Despite our substantial business contingency plans, the impact from natural disasters and other catastrophic events, including terrorist attacks, may have a negative effect on our business and infrastructure, including our information technology systems.	0.9826	0.0089	0.0086
989788	Certain types of losses, generally of a catastrophic nature, such as earthquakes, hurricanes and floods, or terrorist acts, or liabilities that result from breaches in the security of our information systems may be uninsurable or too expensive to justify obtaining insurance. As a result, we may not be successful in obtaining insurance without increases in cost or decreases in coverage levels.	0.9827	0.0088	0.0085
1221704	Certain of the Company's properties are located in California or in other areas with higher risk of earthquakes. In addition, many of the Company's properties are located in coastal regions, and would therefore be affected by any future increases in sea levels or in the frequency or severity of hurricanes and tropical storms, whether such increases are caused by global climate changes or other factors.	0.9826	0.0089	0.0085
1362441	Such activities may also adversely affect the Company's financial condition and results of operations by reducing consumer confidence in the marketplace and by modifying consumer spending habits. Severe weather, natural disasters and adverse climate changes may adversely affect the Company's financial condition and results of operations.	0.9828	0.0088	0.0085
Cpy ID	Text passages not related to climate risk	phys	trans	other
152905	This instrumentation is used in either static or mobile laboratories. Customers include local, state, federal, and international law enforcement agencies and health laboratories.	0.0131	0.0154	0.9715
47928	In 2015, the Company commenced development of a two-phase multi-family development of the CitySquare project in Worcester, Massachusetts. The first phase, with 237 units, is under construction with anticipated initial deliveries in the fourth quarter 2017.	0.0133	0.0158	0.9709
855198	The Company has contracts to acquire raw base paper from certified photographic paper suppliers during the next several years. Lithographic aluminum is the primary material used in the manufacture of offset printing plates.	0.0145	0.0189	0.9666
904659	Competition for homebuyers could reduce our deliveries or decrease our profitability. The homebuilding industry is highly competitive, not only for skilled labor, material and suitable land, as described in previous risk factors, but also for new homebuyers.	0.0134	0.0152	0.9715
1119061	The global economic downturn that occurred in 2008 and 2009, coupled with the global financial and credit market disruptions, had a historic negative impact on the aluminum industry and Alcoa. These events contributed to an unprecedented decline (56% in the last five months of 2008) in LME-based aluminum prices, weak end markets, a sharp drop in demand, increased global inventories, and higher costs of borrowing and/or diminished credit availability.	0.0136	0.0154	0.9710

data through the Datastream Navigator tool. Once the navigator is opened, we click on "Explore" and go through the following steps in the explorer:

1. Select "Credit Default Swaps" (twice)
2. Select "Thomson Reuters CDS by Tenor"
3. Select "No Restructure" (CDS clause - instrument-codes ending on "AX", this clause is chosen simply because it contains the most instruments)
4. Select wanted "Tenor" (All different maturities are done separately)

Once this selection is made, all of the different available CDS contracts appear. Before downloading this information, we apply the following filters:

- Currency: US Dollar denoted contracts only
- Market: United States only

Once these steps and filters have been applied all remaining instruments are selected and downloaded. We repeat the above steps for all available maturities. Daily time series have been downloaded from 2009 till 2021. Note that although not all instruments have data for the entire period, this is not a problem. For the period for which that instrument has full data, the instrument is kept. Finally, before ending up with our final CDS dataset, instruments not complying with the standard Datastream mnemonic codes are also excluded, such as those not ending on "AX" or having the wrong order. Negative spreads are also excluded. We now observe daily CDS spreads for 767 instruments for maturities of ranging from six months to 30 years.

To gain some overview of our dataset, we provide some descriptive statistics of the independent variables and the CDS spreads. In Table 2, we report the corresponding mean, standard deviation, skewness, and excess kurtosis together with the minimum, median, and maximum values of the different variables.

Table 2. Descriptive statistics

Descriptive statistics of all independent variables, five-year CDS spreads and the five-year minus one-year CDS term structure slope. *Transition* and *Physical* are climate risk scores based on our BERT algorithm. S^{5Y} (in basis points) is the five-year CDS spread. $Y^{5Y,1Y}$ (in basis points) is the five-year minus one-year CDS term structure slope. *Lev* (in percentage points) is the leverage ratio, defined as ratio between the book value of total liabilities and the sum of the book value of total liabilities and the market value of equity. *ROA* (in percentage points) is the return-over-assets, defined as the ratio of net income to total assets. *Vol* (in percentage points) is the daily return volatility over the last 180 trading days. *IR* (in percentage points) is the interest rate. *BC* (in percentage points) is the return on the S&P500 index. The table presents the average value (Mean), standard deviation (Stdv), skewness (Skew) and excess kurtosis (Ex. kurt), as well as the minimum (Min), maximum (Max) and median (Median) values for each variable. The sample period ranges from February 2010 to October 2021.

	Mean	Stdv	Skew	Ex. Kurt	Min	Median	Max
Physical	0.03	0.04	2.98	23.43	0.00	0.02	0.58
Transition	0.06	0.10	2.07	4.19	0.00	0.02	0.67
S^{5Y} (bp)	166.09	190.93	3.20	14.02	8.85	98.13	1979.88
Lev (%)	47.29	21.61	0.51	-0.58	0.86	43.95	99.66
ROA (%)	1.17	2.59	-4.67	201.61	-96.08	1.07	39.39
Vol (%)	1.93	1.05	2.51	11.88	0.19	1.64	14.43
IR (%)	1.01	3.94	-0.53	1.18	-13.37	1.70	11.94
BC (%)	2.22	0.68	-0.13	-0.01	0.62	2.21	3.85

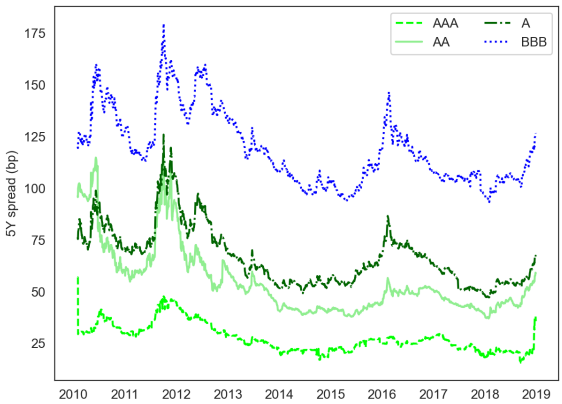
In Panels A and B of Figure 5, we plot the average five-year CDS spreads over the entire sample period for different rating categories. Spread levels are at their highest levels at the end of 2011, following a steep jump upwards. After this spike, the spreads slowly start to decrease, with an intermediate bump in 2016 and towards the end of the sample period. In Panels C and D, we also plot the average CDS spread term structure per rating category. In line with the previous literature, we find that the average term structure of CDS spreads is upward-sloping, except for low-quality firms. A positively sloped term structure reflects the intuition that a firm's credit quality declines over time, resulting in higher costs of default protection and CDS spreads. We find that the average term structure usually flattens out at a time-horizon of 10 years.

We note that the average CDS term structures are typically upward sloping, except for the worst rating category. This observation is in line with Helwege and Turner (1999). Hence, the credit models outlined in Section 2 of the paper cannot capture this feature of the data. To remedy this model deficiency, Collin-Dufresne and Goldstein (2001) assume a stationary leverage ratio so that the default boundary migrates upward over time along with firm value. However, we do not consider

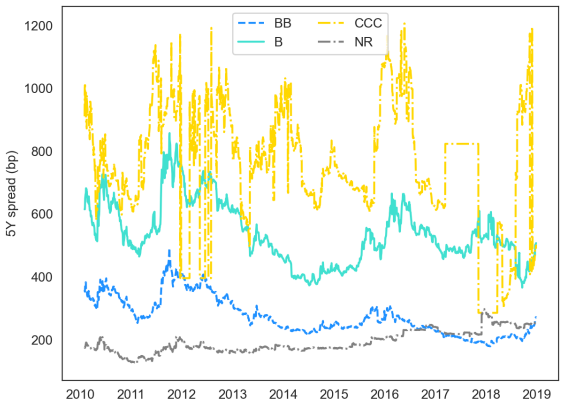
Figure 5. Average CDS spreads and term structure

For the different rating classes, panel (a) and (b) present average five-year CDS spreads, while panel (c) and (d) present average term structures. The sample period ranges from February 2010 to December 2018.

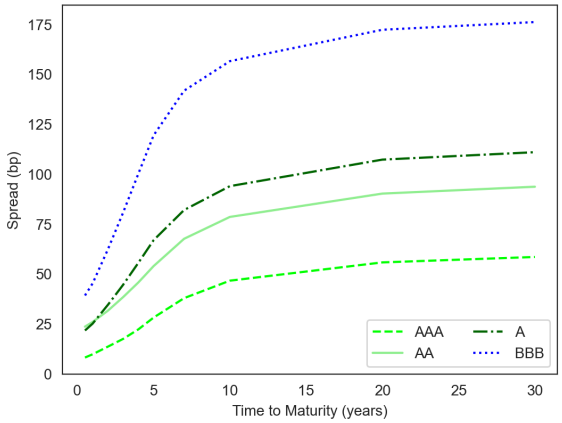
Panel A: Investment - 5Y spread



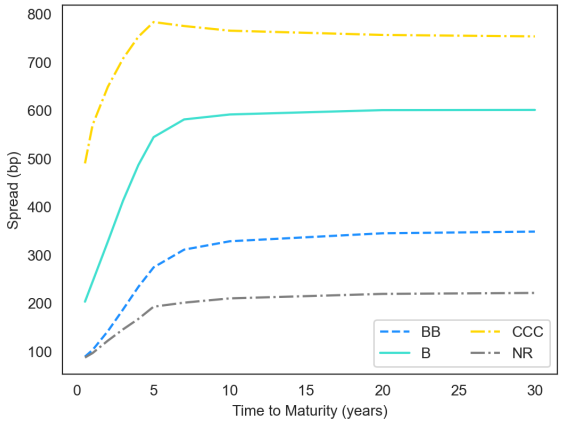
Panel B: Non-Investment - 5Y spread



Panel C: Average term structure Investment



Panel D: Average term structure Non-Investment



such an extension for our analysis, since we are only interested in the qualitative prediction of the impact of credit risk disclosure.

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